

Basic Principles of Medicine 2
Module : Digestion and Metabolism
Lecture No: 18

Title: Adipose tissue lipolysis; Fatty acid oxidation

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Objectives: Adipose tissue lipolysis, Fatty acid oxidation

SOM. MK.I.BPM2.3.DM.2.BCHM.1193	Describe adipose tissue lipolysis and the role of hormone sensitive lipase.
SOM. MK.I.BPM2.3.DM.2.BCHM.1194	Describe fatty acid activation
SOM. MK.I.BPM2.3.DM.2.BCHM.1195	Describe the carnitine shuttle mechanism.
SOM. MK.I.BPM2.3.DM.2.BCHM.1196	Explain the consequences of defects in transport of fatty acids (carnitine deficiency and CPT-I and CPT-II deficiency)
SOM. MK.I.BPM2.3.DM.2.BCHM.1197	List the reactions of the mitochondrial β -oxidation pathway (not substrate names)
SOM. MK.I.BPM2.3.DM.2.BCHM.1198	Outline the energetics of β -oxidation
SOM. MK.I.BPM2.3.DM.2.BCHM.1199	Indicate the end products of oxidation of odd chain fatty acids
SOM. MK.I.BPM2.3.DM.2.BCHM.1200	Outline alpha-oxidation of branched chain fatty acids and indicate the metabolic defect in Refsum disease.
SOM. MK.I.BPM2.3.DM.2.BCHM.1201	Review the pathogenetic mechanism and metabolic changes in Jamaican vomiting sickness
SOM. MK.I.BPM2.3.DM.2.BCHM.1202	Compare and contrast systemic beta-oxidation defects and myopathic beta oxidation defects. Identify causes for each.

Recommended Reading

- Lippincott's Biochemistry 9th Edition: Chapter 16
 <https://sgu.idm.oclc.org/login?url=https://premiumbasicsciences.lwwhealthlibrary.com/book.aspx?bookid=3402>
- Reference: Lippincott's Biochemistry Lippincott's Chap 16,pg 189-200;
Concept map Fig 16.25; [Fatty Acid, Triacylglycerol, and Ketone Body Metabolism | Lippincott Illustrated Reviews: Biochemistry, 9e | Premium Basic Sciences | Health Library](#)
- [Fatty Acid Oxidation Disorders: Physiology | Premium Basic Sciences | Health Library](#)
- Lippincott's Table Fig. 28.29 (Vitamins reference)
- **Lippincott's Questions: 16.1 – 16.5**

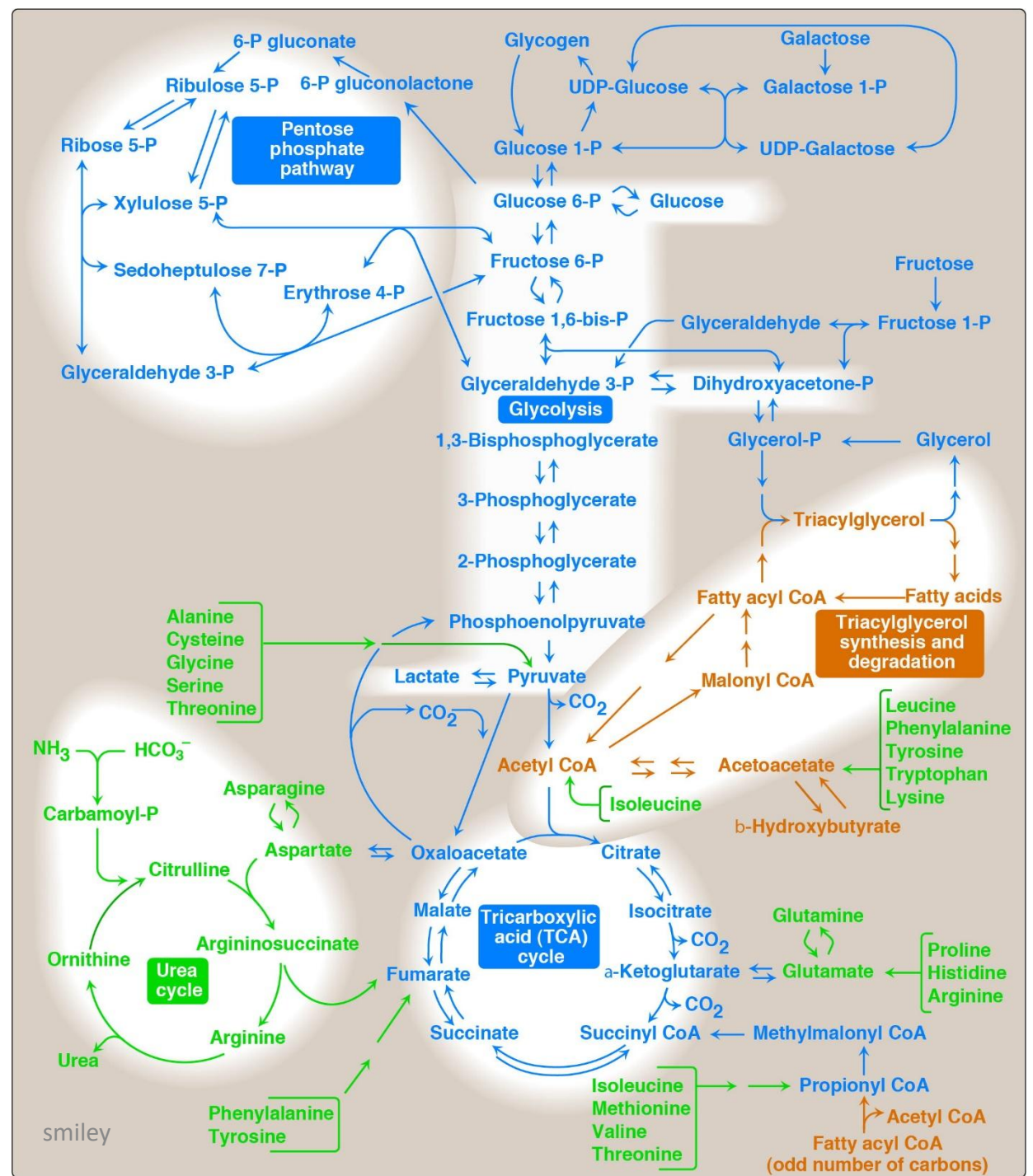
Intermediary metabolism (Metabolic map)

DLA: Overview of metabolism in the fed state and during fasting

Purpose of pathway: Liver and muscle
Regulation: Fed vs fasting
Disorders

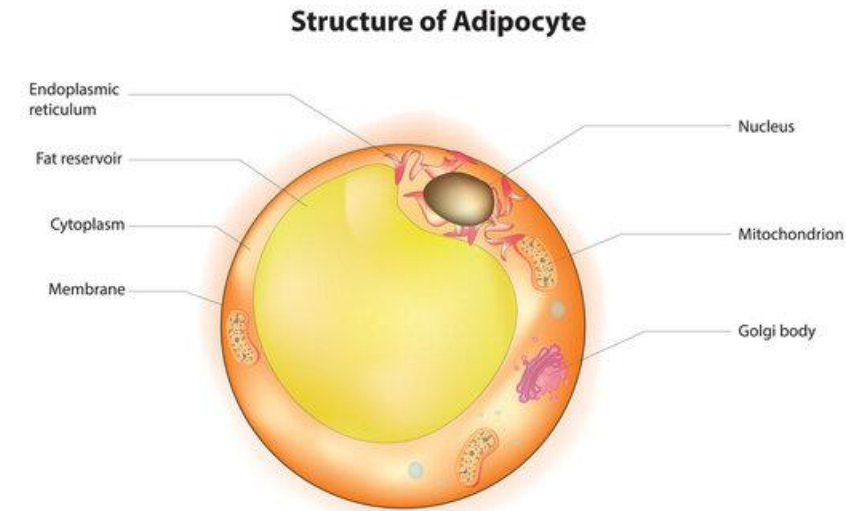
SOM. MK.I. BPM2.3. DM.2.BCHM.1089

Lippincott's figure 8.2

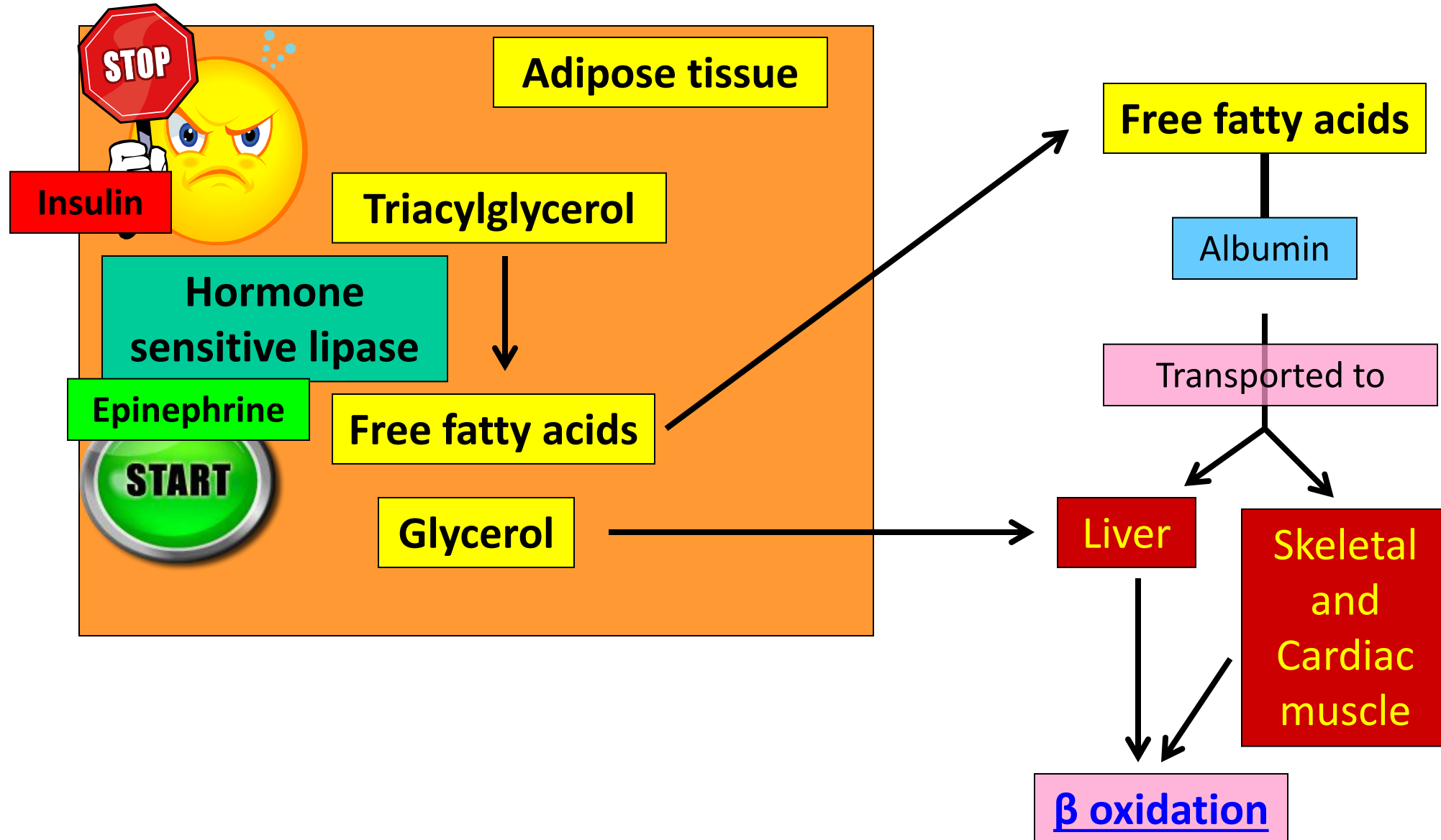


Lipid mobilization from adipose tissue (Lipolysis)

- Triacylglycerol (TAG) is storage lipid
 - Lipid droplets in **white adipose tissue**
- Fatty acids in TAG released as free fatty acids (non-esterified fatty acids) by HS Lipase
- Glycerol formed by lipolysis, not reused in adipose tissue (adipocytes lack glycerol kinase)
 - Glycerol goes to liver, to enter glycolysis or gluconeogenesis or TAG synthesis
- **Hormone sensitive (HS) lipase**
 - Inhibited by insulin (inactive on dephosphorylation)
 - Low levels of insulin (fasting), stimulates HS lipase
 - Epinephrine stimulates HS lipase (Phosphorylated active)



Lipid mobilization from adipose tissue

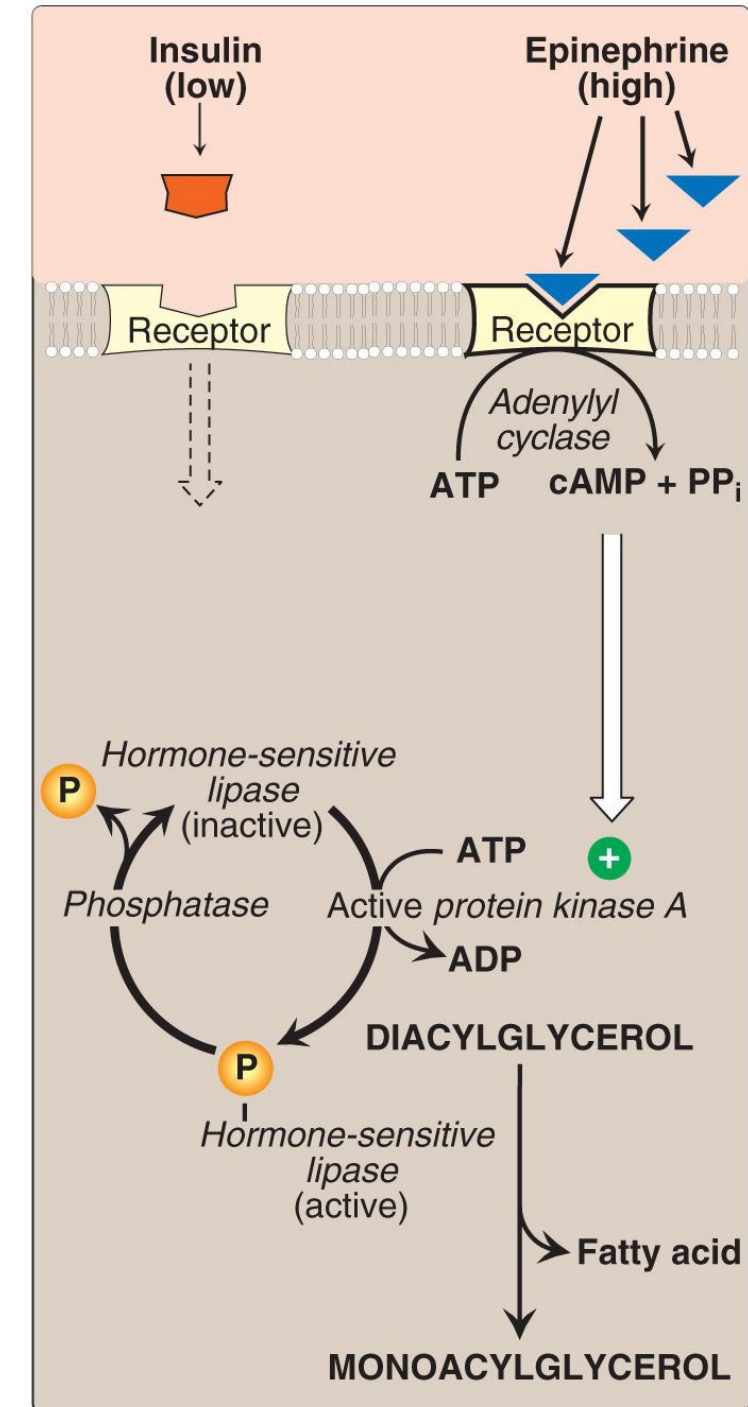


Role of Epinephrine: Fasting, flight and fight

- Activates protein kinase A
- Phosphorylation of HS Lipase → Active form
 - Stimulates hydrolysis of TAG to free fatty acids and glycerol

Insulin resistance and Diabetes mellitus:

- Explain changes in type 1 and type 2 diabetic
- Elevated serum free fatty acids: Indicator of insulin resistance



Compare: Hormone sensitive lipase and Lipoprotein lipase

Self - study

Hormone sensitive lipase vs Lipoprotein lipase vs Pancreatic lipase

Hormone sensitive lipase

- Location: Within adipocytes
- Role of insulin: Insulin inhibits HS lipase
- Action: Breaks down storage TAG in adipose tissue to free fatty acids and glycerol
- Lab marker for increased HS lipase activity: Elevated serum free fatty acids
- Increased HS lipase activity seen in
 - Starvation
 - Insulin resistance (Metabolic syndrome)
 - Uncontrolled type 1 diabetes mellitus – HS lipase activity markedly increased

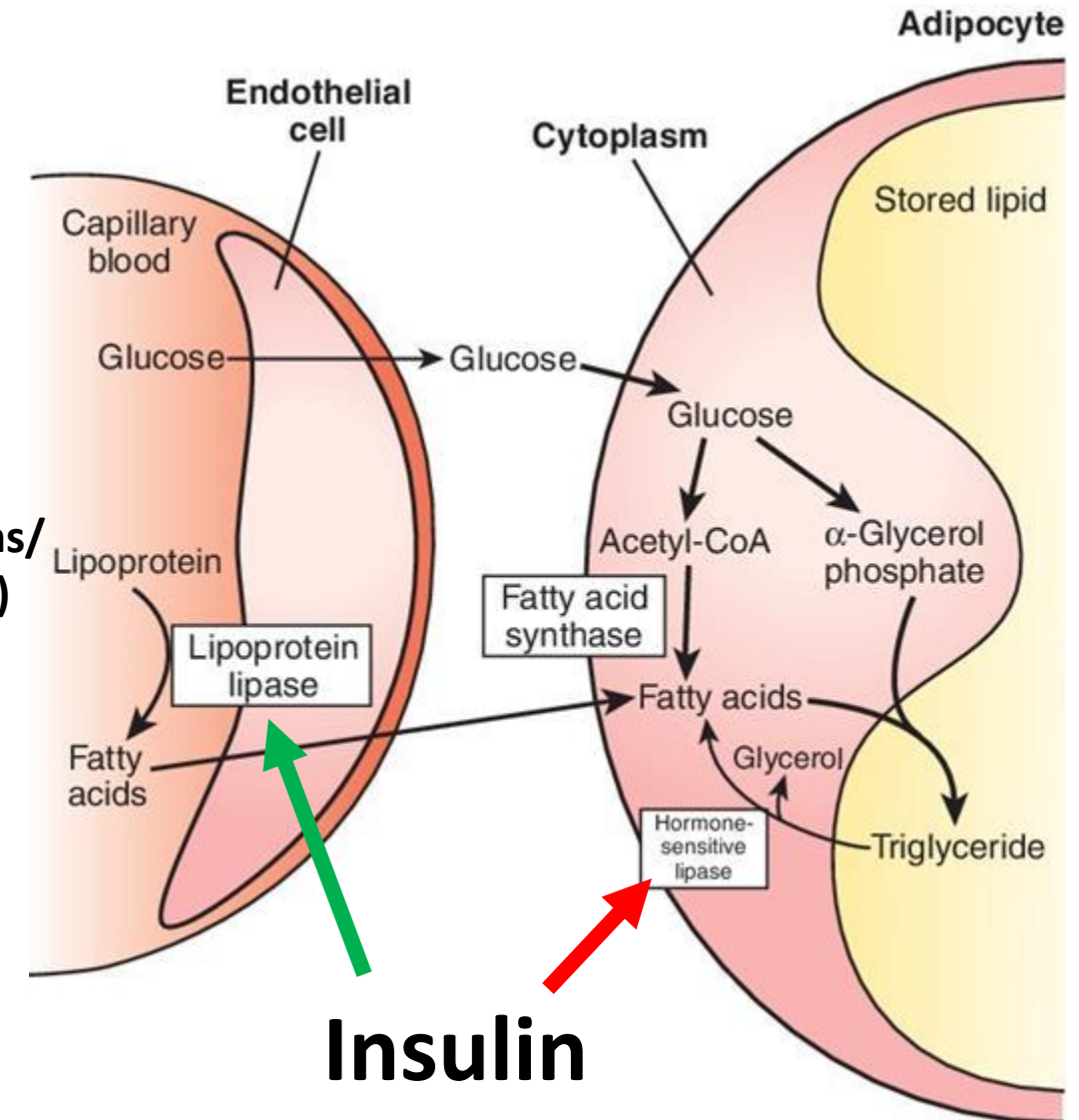
Lipoprotein lipase

- Location: Endothelium of blood vessels in adipose tissue and muscle
- Insulin induces lipoprotein lipase and required for optimum activity
- Action: breaks down TAG in lipoproteins (chylomicrons and VLDL) to free fatty acids that enter adipocytes and form TAG for **storage within adipocytes**
- Lab marker for low lipoprotein lipase activity: Increased TAG; and Increased VLDL
- Decreased activity of lipoprotein lipase is seen in
 - Insulin resistance (Metabolic syndrome)
 - type 2 diabetes mellitus – this results in increased serum TAG (VLDL) – which in turn increases the risk for cardiovascular disease (macrovascular complications)

Pancreatic lipase: Enzyme for digestion of dietary TAG

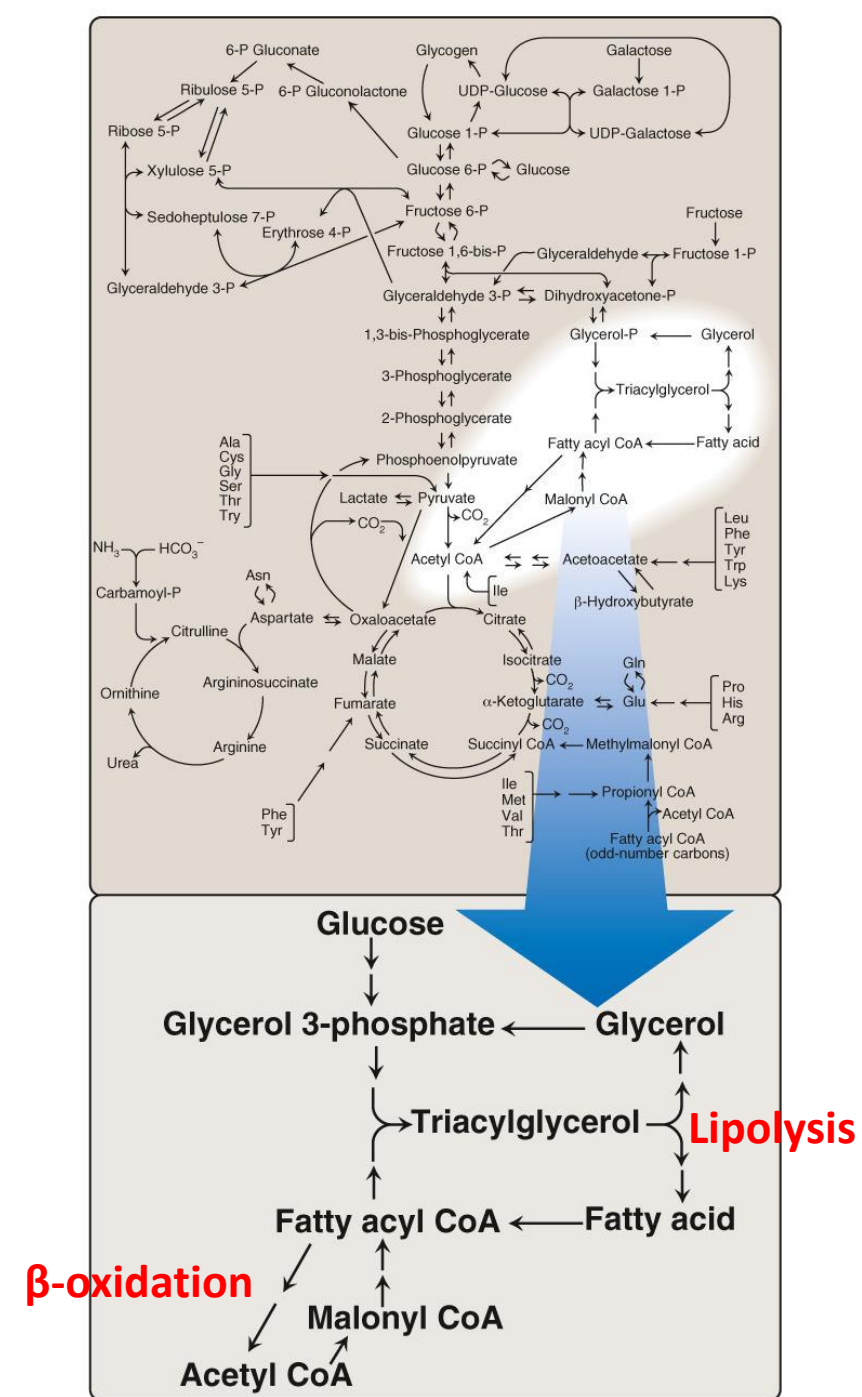
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Chylomicrons/
VLDL (TAG)



Overview of fatty acid oxidation

- Free fatty acids bound to albumin in circulation
- Free fatty acids transported to **liver and muscle (skeletal and cardiac muscle)** – sites of β -oxidation **(AEROBIC)**
- Fatty acids NOT oxidized by brain; NOT important fuel for brain even during prolonged starvation
- Fatty acids cannot be oxidized by cells that lack mitochondria (RBCs) or oxygen



β -oxidation

- Oxidation of fatty acids at β -carbon atom of fatty acid
- Intracellular location: Mitochondria
- Stages of β -oxidation
 - Activation of fatty acid (cytosol)
 - **Transport of fatty acid from cytosol to mitochondria (carnitine shuttle)**
 - β -oxidation proper (reactions of β -oxidation)

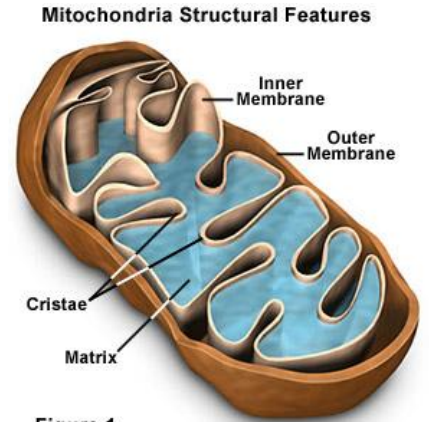
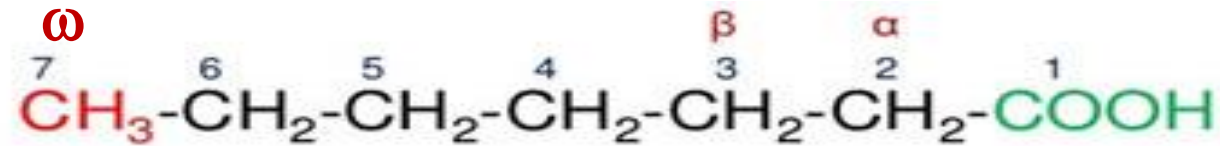
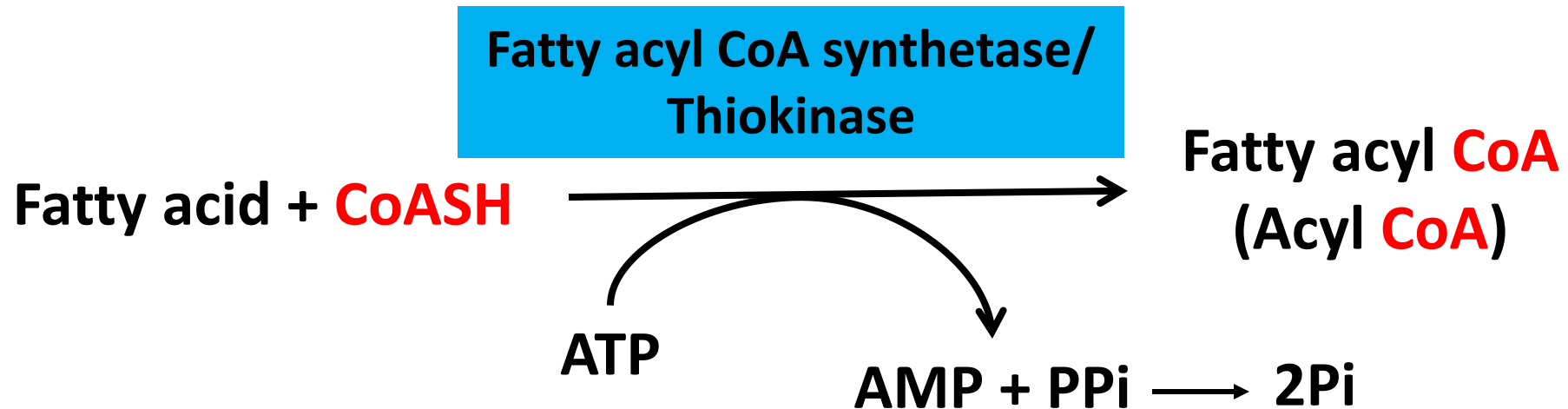


Figure 1



Activation of fatty acid

- Activated to fatty acyl CoA by Fatty acyl CoA synthetase (thiokinase)
- Present in outer mitochondrial membrane (cytosolic side)



Transport of fatty acid from cytosol to mitochondria (Carnitine shuttle)

SOM. MK.I.BPM2.3.DM.2.BCHM.1195

Mitochondria Structural Features

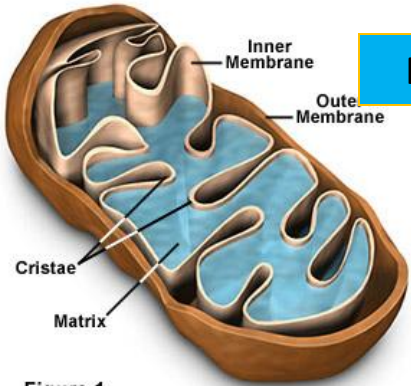
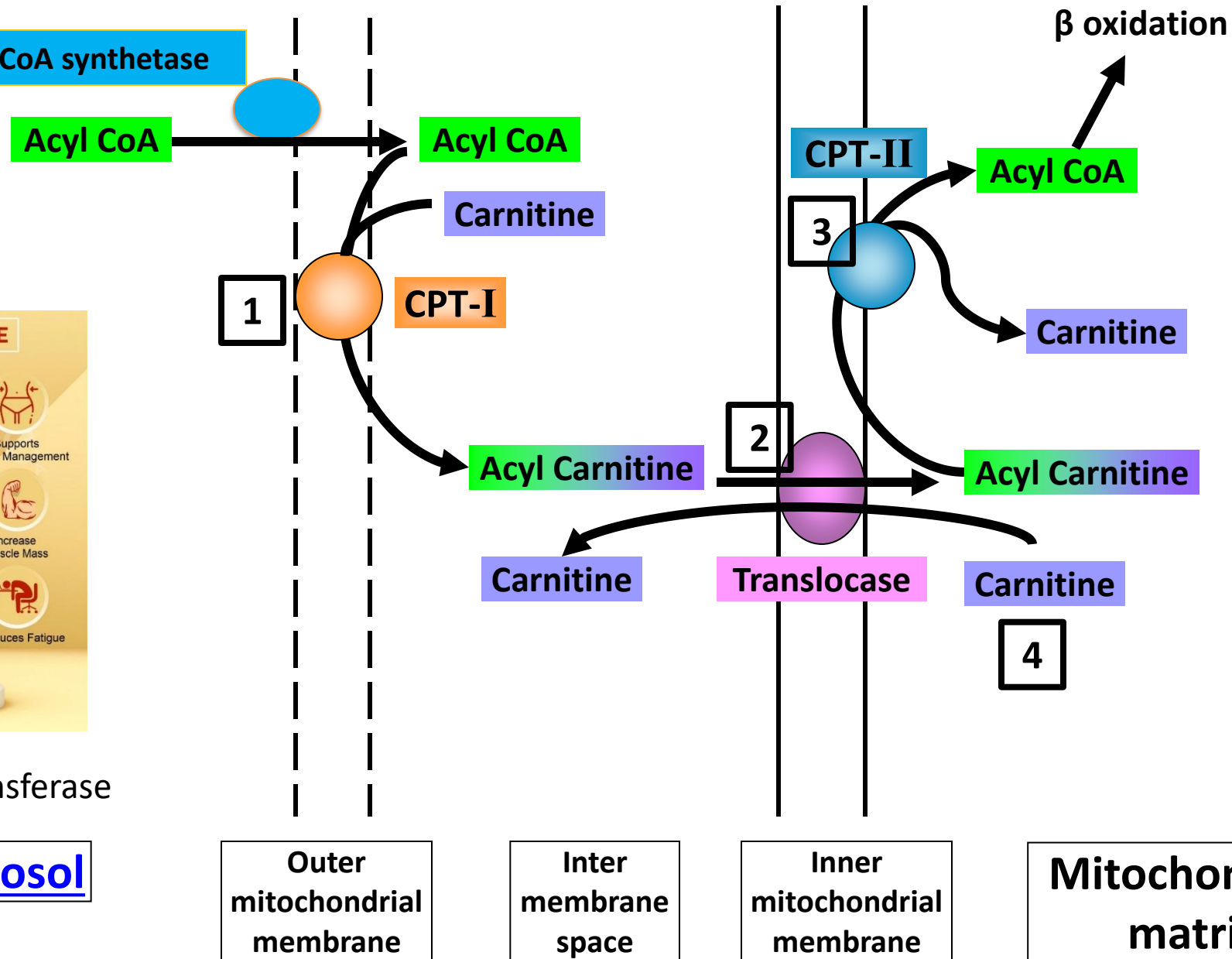


Figure 1



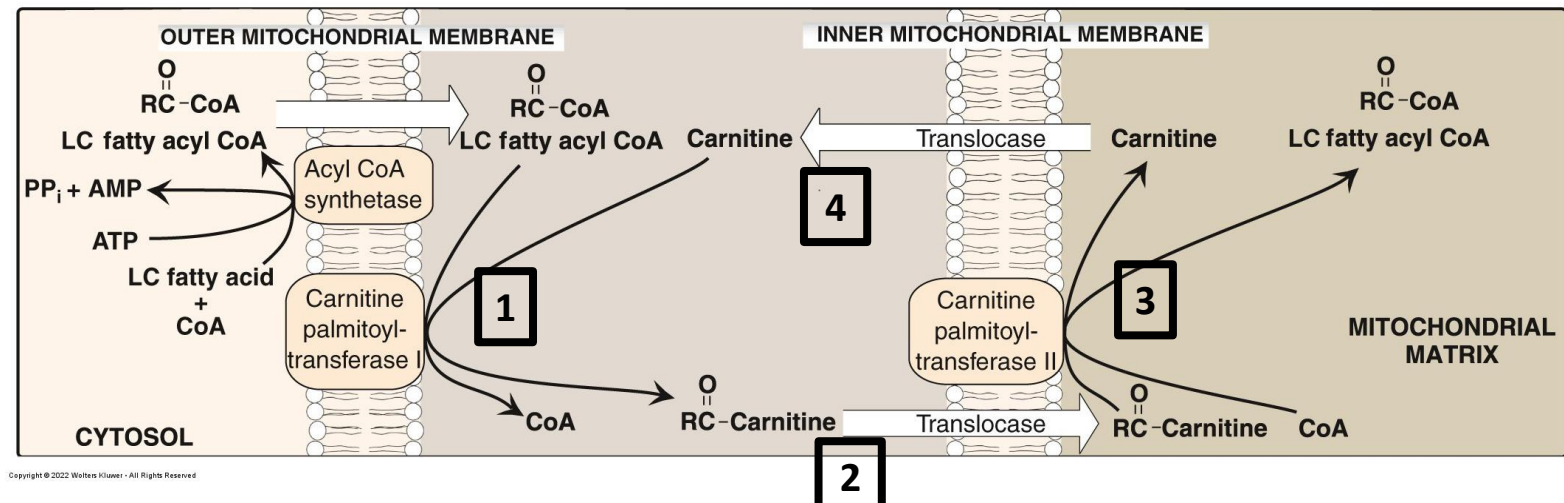
CPT: Carnitine palmitoyl transferase or carnitine acyl transferase

Cytosol



Transport of fatty acid from cytosol to mitochondria (**Carnitine shuttle**)

- CPT-I and CPT-II in outer and inner mitochondrial membrane respectively
- CPT-I and CPT-II isoforms in liver and muscle
- Fatty Acyl CoA cannot travel through inner mitochondrial membrane
 1. **Carnitine** binds acyl group to form acyl-carnitine (CPT-I)
 2. Acyl carnitine transported across inner mitochondrial membrane via translocase
 3. Acyl CoA formed in matrix (CPT-II) and used for β -oxidation
 4. **Carnitine** transported back to intermembrane space via translocase
- **Regulation: Malonyl CoA inhibits** CPT-I (when fatty acid synthesis is active, β -oxidation inhibited)

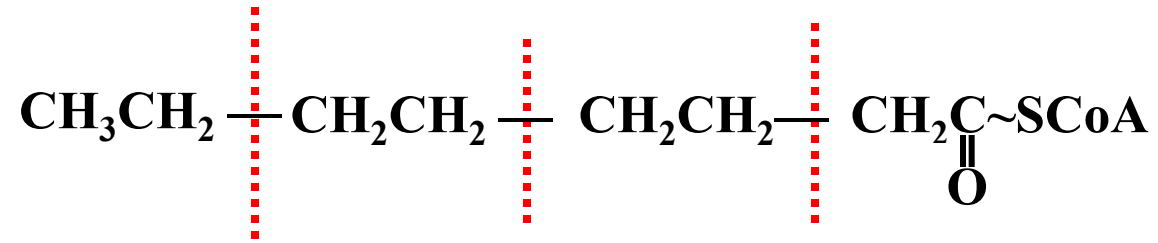


Reactions of β -oxidation

- Enzymes in mitochondrial matrix
- Sequence of reactions of β -oxidation
 - Oxidation (removal of H) (requires FAD)
 - Addition of water
 - Oxidation (removal of H) (requires NAD^+)
 - Cleavage
- One cycle of β -oxidation: sequence of four reactions
- Results in cleavage of 2 C-atoms (removed as acetyl CoA)

Overview of Beta Oxidation

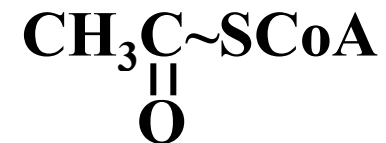
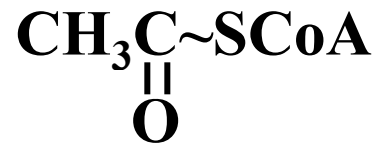
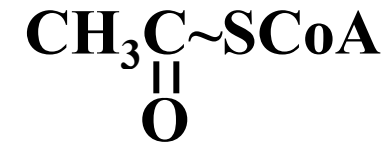
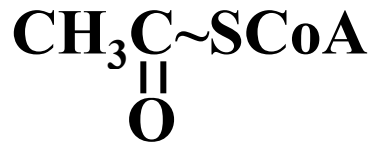
8 carbon Fatty Acid Acyl-CoA



4 two carbon Acetyl-CoAs

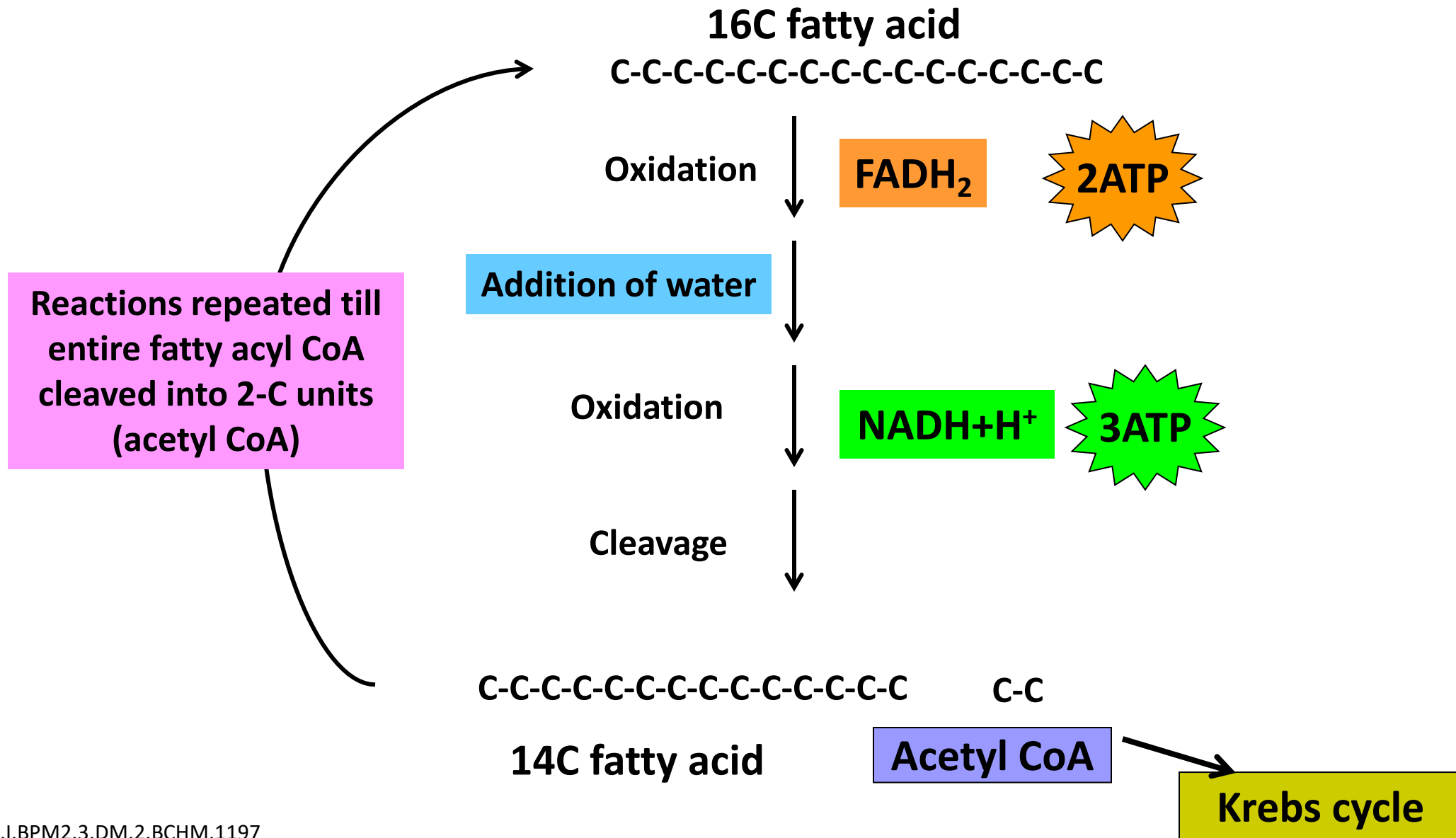
Overview of Beta Oxidation

8 carbon Fatty Acid (Acyl-CoA)



4 two carbon Acetyl-CoAs

Reactions of β -oxidation

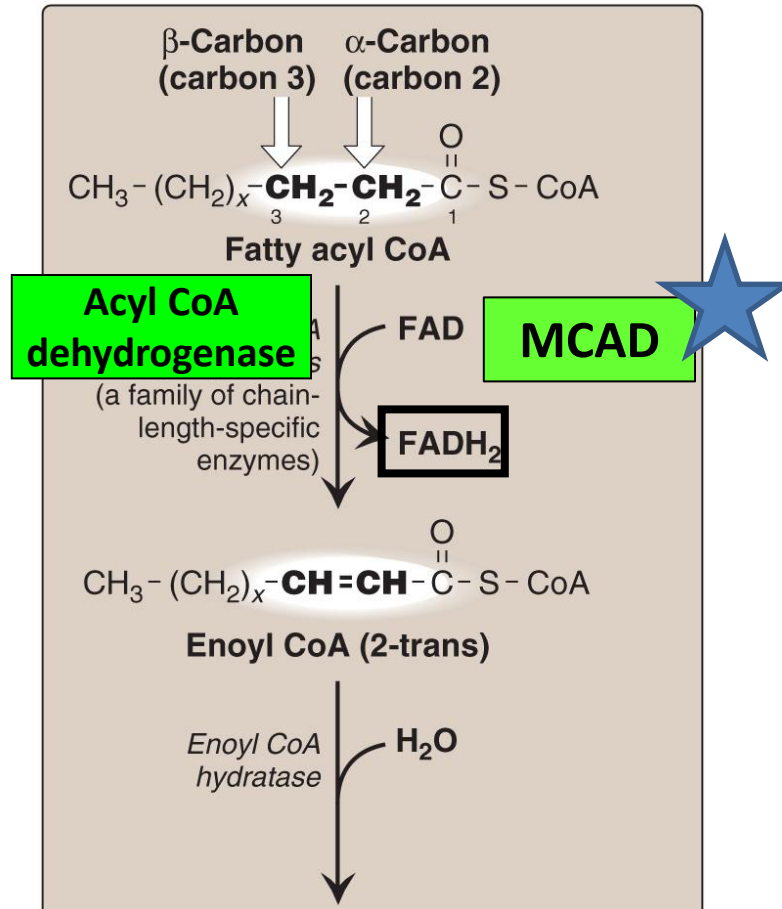


Enzymes of β -oxidation

- **Step 1- Acyl CoA dehydrogenase** (requires FAD) - Oxidation
- Step 2- Enoyl CoA hydratase – Addition of water
- Step 3- 3-hydroxy acyl CoA dehydrogenase (requires NAD^+) - Oxidation
- Step 4- Thiolase - Cleavage

Reactions of β -oxidation

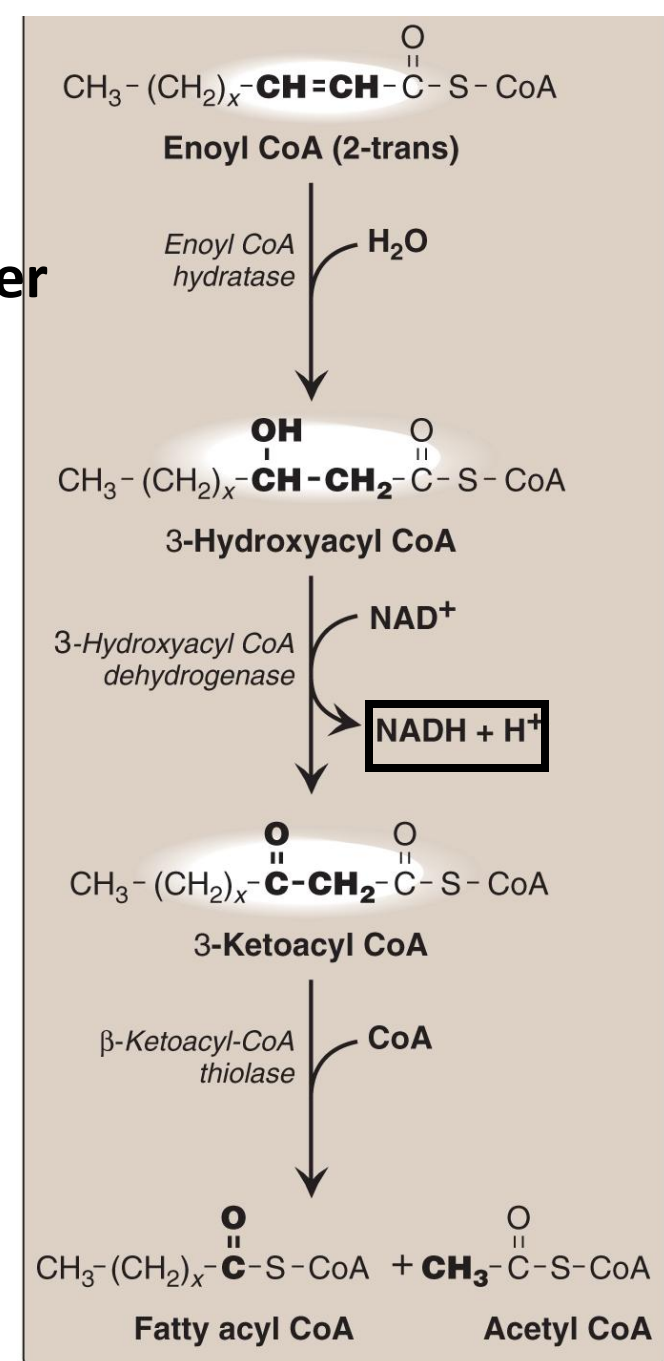
Oxidation



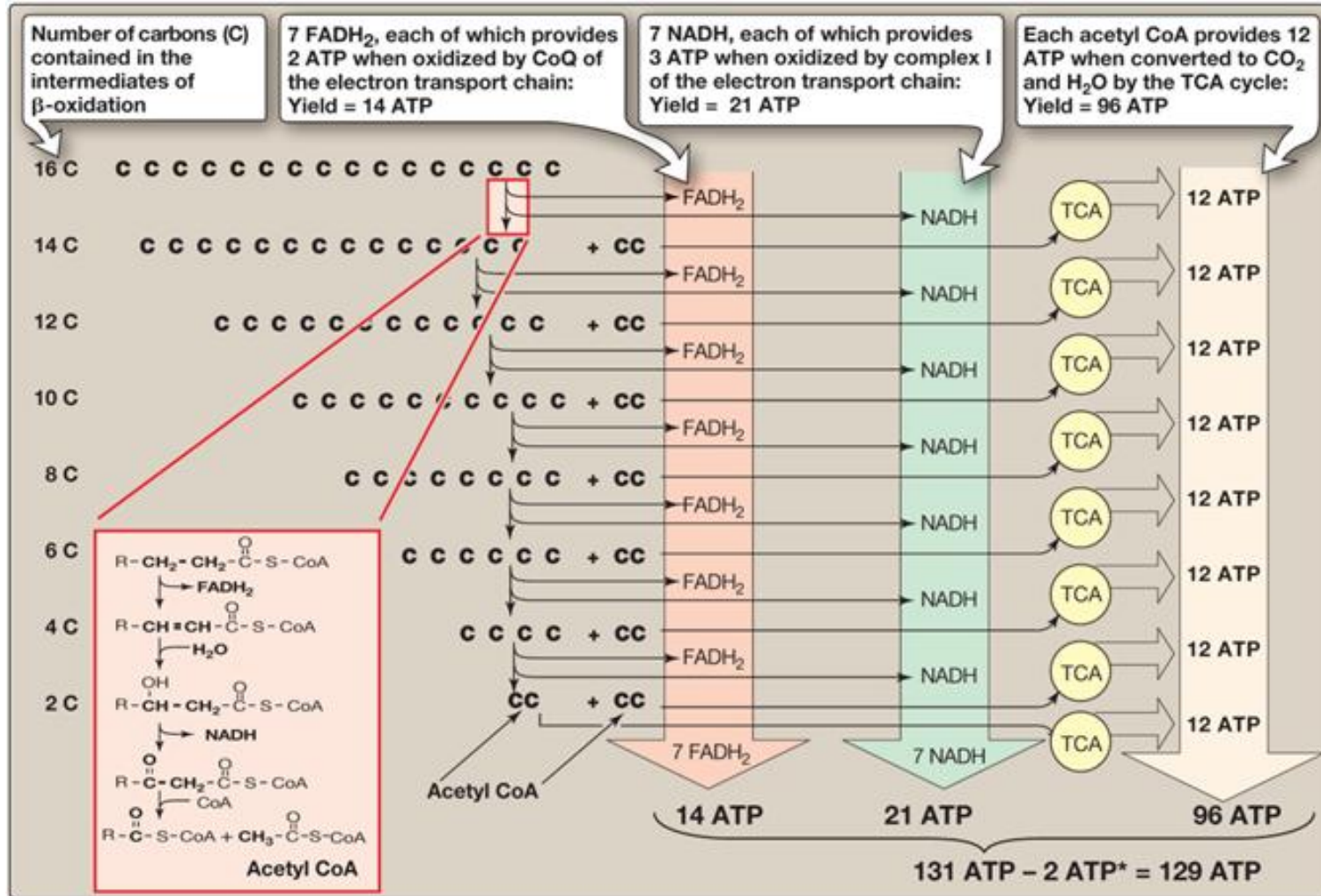
Addition of water

Oxidation

Cleavage



Reactions of β -oxidation on Palmitoyl CoA

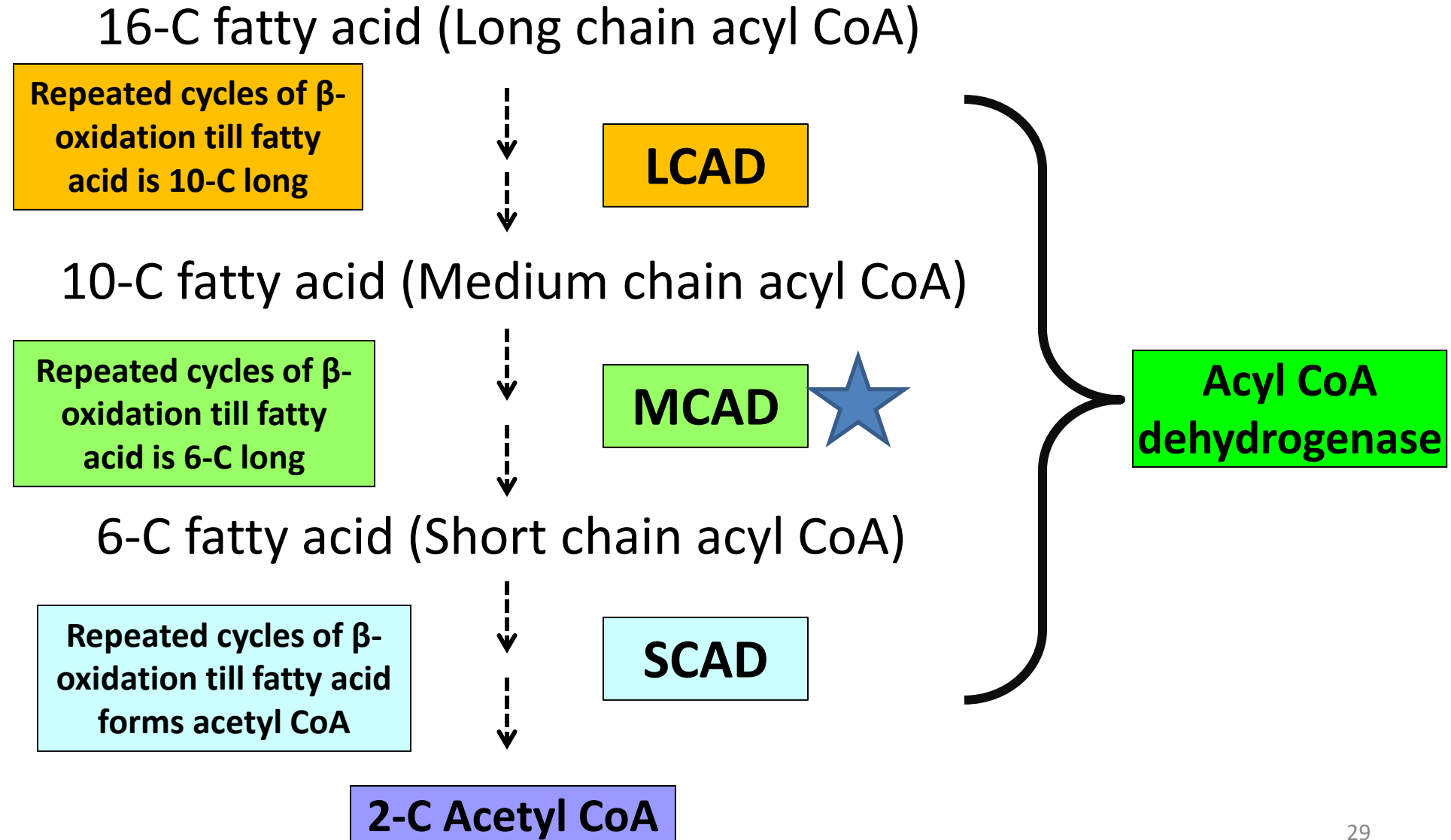


- One cycle of β -oxidation: Acetyl CoA, FADH_2 and $\text{NADH}+\text{H}^+$
- Complete β -oxidation of 16C palmitic acid results **8 Acetyl CoA, 7 FADH_2 , 7 $\text{NADH}+\text{H}^+$**
- Acetyl CoA enters TCA cycle to form 12 ATP

Energy yield from β -oxidation of palmitic acid

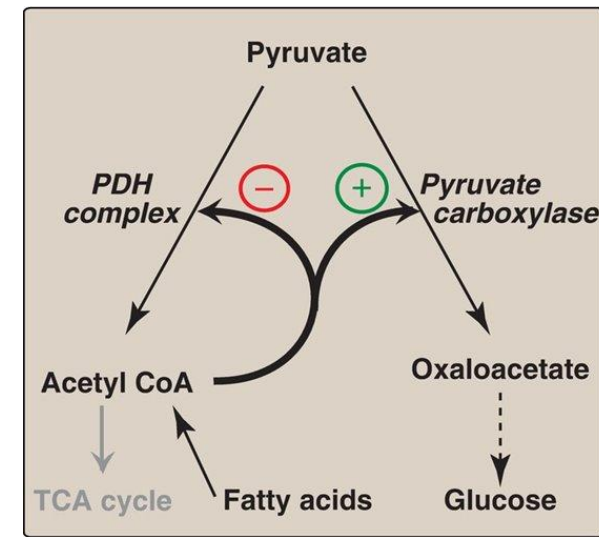
- β -oxidation of 16C palmitic acid = 8 Acetyl CoA, 7FADH₂, 7NADH+H⁺
- 8 Acetyl CoA x 12 ATP (TCA cycle) = 96 ATP
- 7FADH₂ x 2 ATP = 14 ATP
- 7NADH+H⁺ x 3 ATP = 21 ATP
- Total = 131 ATP
- **-2 ATP** for activation of fatty acid (Fatty acyl CoA synthetase)
- Net ATP formed = **129 ATP** (Also refer Fig 16.18 – Lippincott's)
- Compare to oxidation of glucose (38 ATP per glucose molecule)
- (1NADH=2.5ATP and 1FADH₂=1.5ATP; We use 1NADH =3ATP and 1FADH₂=2ATP for ease of calculation)

Beta-Oxidation palmitic acid: **Acyl CoA dehydrogenase**: LCAD, **MCAD** and SCAD



Fatty acid oxidation in Liver

- Overnight fast, 60-70% energy from fatty acid oxidation
- Beta-oxidation in **liver** is important for
 - **Gluconeogenesis**: Energy and acetyl CoA (activator of pyruvate carboxylase)
 - **Ketone body synthesis**: Acetyl CoA for ketogenesis
- **Systemic fatty acid oxidation disorders** (MCAD deficiency, primary carnitine deficiency, CPT-I deficiency): Affects liver
 - **Hypoglycemia** (Cells rely on glucose for energy and reduced gluconeogenesis) – Acetyl CoA absolute activator for pyruvate carboxylase (gluconeogenesis)
 - **Hypoketosis** (reduced formation of acetyl CoA)

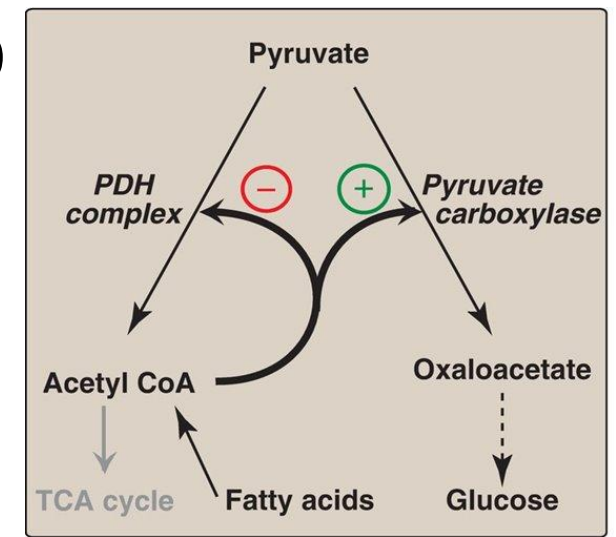


Case report: MCAD deficiency

- An 18-month-old girl brought to emergency room because she became unresponsive. <https://www.youtube.com/watch?v=p9APKkv8RC0>
- Episodes of **hypoglycemia** ever since she was 11 months old
- Episodes occurred on fasting (>8 hours) or **following illness**
- On admission (metabolic crisis),
 - Plasma glucose is 38mg/dL (**Hypoglycemia**)
 - Free fatty acids in blood are elevated (normal), but
 - Ketone bodies are below normal (**Hypoketosis**)
 - C8-C10 acyl carnitines elevated
- **Differentiate from von Gierke disease**

Systemic fatty acid oxidation disorders: **MCAD deficiency**

- **Autosomal recessive disorder (Newborn screen)**
- **Systemic beta-oxidation defect**
- Age of presentation: **6 -24 months**
- **Severe hypoglycemia (metabolic crisis)** on fasting or illness
 - Tissues (liver, muscle) cannot use fatty acids for energy. Glucose is sole source of energy, resulting in profound hypoglycemia
 - **Impaired gluconeogenesis**: Less ATP and acetyl CoA
 - Liver dysfunction and elevated AST and ALT during crisis
- Decreased oxidation of **medium chain fatty acids** (8 -10 C atoms)
 - Less acetyl CoA → Less ketone body synthesis



Systemic fatty acid oxidation disorders: **MCAD deficiency**

- Lab findings:
 - **Hypoglycemia**
 - **Low ketone body levels (hypoketosis)**
 - **Medium chain acylcarnitines (8-10 C)** in urine
 - **Dicarboxylic acids** in urine (Increased **ω -oxidation**)
- Management:
 - Manage **hypoglycemia** using IV glucose
 - Frequent feeding; high carbohydrate, low-fat diet
 - **Avoid fasting; monitor during illness**
 - **Carnitine supplements**

Systemic fatty acid oxidation disorders: **Carnitine deficiency**

- Defect in transport of long chain fatty acids into mitochondria → Defective β -oxidation
- **Primary or systemic carnitine deficiency:** Early age of presentation
 - Impaired Carnitine uptake into tissues (Refer Pg 191 in textbook)
 - **Hypoglycemia** due to impaired gluconeogenesis (Acetyl CoA); And **Hypoketosis**
 - **Liver dysfunction and elevated AST and ALT**
 - **Cardiomyopathy**
- **Secondary carnitine deficiency**
 - Disorders like MCAD → Loss of acylcarnitine in urine
- **Management:**
 - Avoid fasting; monitor during illness
 - Carnitine supplements

Systemic fatty acid oxidation disorders: **CPT-I deficiency**

- Hypoglycemia and hypoketosis
 - Affects liver isoform (**systemic form**)
- Management:
 - Avoid fasting; monitor during illness
 - Carnitine supplements

Compare liver glycogen storage disorder and systemic fatty acid oxidation defect

Systemic fatty acid oxidation defect

- Hypoglycemia and hypoketosis
- Age: 6 months-2years
- **Liver biopsy: Excessive lipid (TAG) droplets**
- Elevated liver enzymes (ALT and AST)
- MCAD/CPT-I deficiency/ Primary carnitine deficiency

Liver Glycogen storage disorder (von Gierke disease)

- Hypoglycemia
- Ketone body levels (increased)
- Age: 6 months-2years
- **Lactic acidosis**
- **Hypertriglyceridemia; hyperuricemia**
- **Hepatomegaly**
- **Liver biopsy: Excessive Glycogen accumulation (structure normal)**
- Elevated liver enzymes (ALT and AST)
- 'Doll' face
- Glucose 6-phosphatase deficiency

Fatty acid oxidation in skeletal muscle

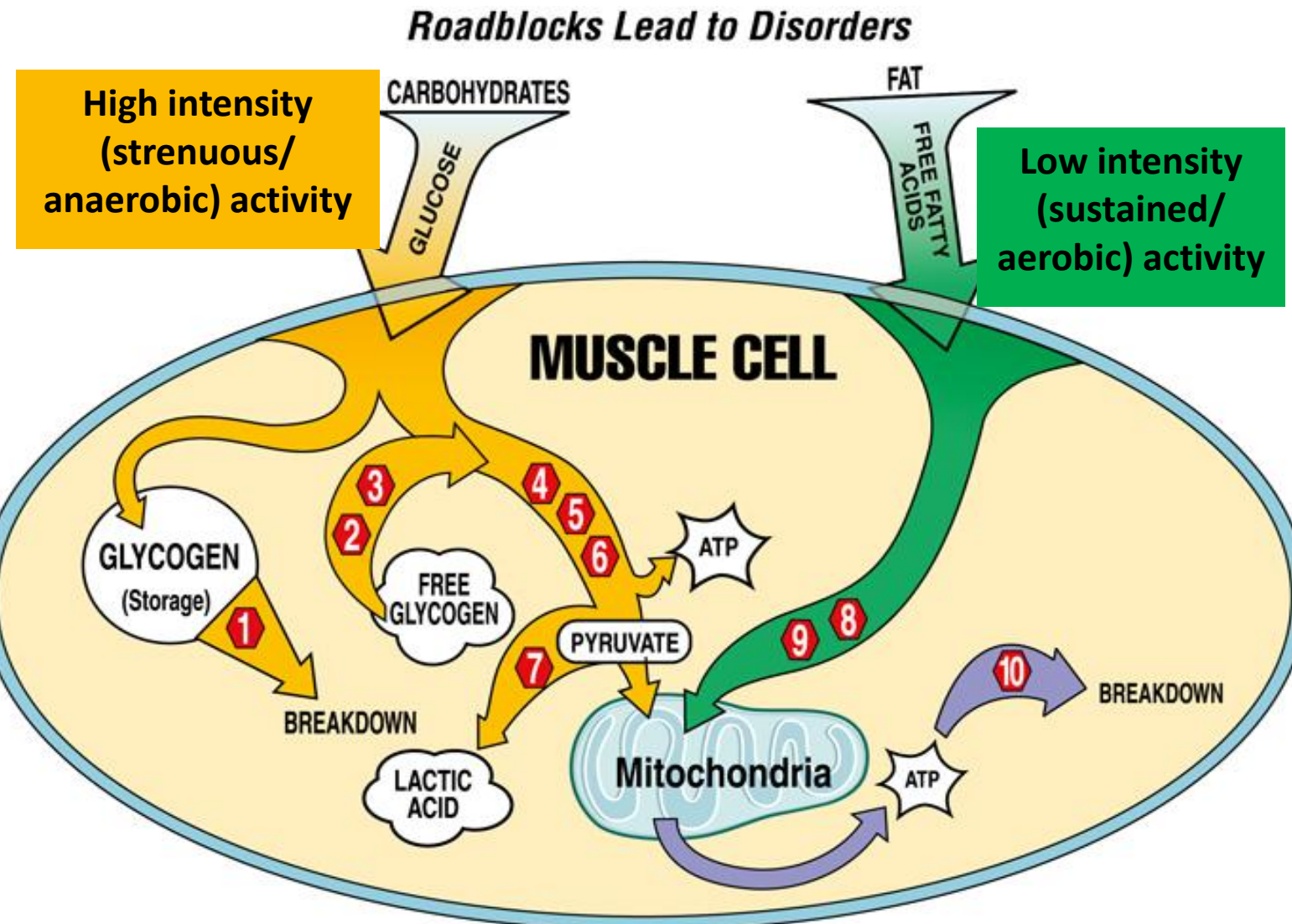
- Skeletal muscle uses fatty acid oxidation during **low-intensity prolonged activity (Endurance or Aerobic)**
- Provide energy for **Aerobic Exercise**
- **Myopathic fatty acid oxidation disorders:** Defect in fatty acid oxidation restricted to muscles (Skeletal and cardiac)
 - **Manifests as Cramps** during low-intensity, sustained, aerobic activity (compare to muscle glycogen storage disorder)
 - **Unable** to perform **aerobic exercise**
- Lab findings:
 - Forearm exercise test: Lactate levels increase; Blood lactate levels rise following ischemic exercise (Indicates normal anaerobic metabolism)
 - Myoglobinuria, and elevated levels of serum CK-MM;
 - Muscle biopsy shows: **Lipid droplets in muscle**

Myopathic fatty acid oxidation disorders: **CPT-II deficiency**

- Cardiomyopathy and muscle weakness (**myopathic form**)
 - Unable to tolerate aerobic activity (**Cramps** during aerobic exercise)
 - **Lipid deposits (triglycerides)** in skeletal muscle
 - Forearm exercise test: Lactate levels increase
 - Myoglobinuria and elevated serum CK-MM levels
 - Affects muscle isoform
- Management:
 - Cease muscle activity and give glucose
 - **Carnitine supplements**

Metabolic myopathies

<https://www.mda.org/disease/metabolic-myopathies/types>



Muscle glycogen storage disorder

Muscle phosphorylase deficiency (McArdle)

Phosphofructokinase deficiency (Tarui)

Myopathic fatty acid oxidation defect

Carnitine palmitoyl transferase-II deficiency

Skeletal muscle depends on energy from carbohydrates and fats. Fuels stored in muscle (glycogen) or imported directly from bloodstream (glucose and fatty acids)

Differentiate between muscle glycogen storage disorder and myopathic fatty acid oxidation defect

Aerobic and anerobic exercise: Fuel sources

Aerobic exercise



- Oxygen present; Abundant mitochondria
- Glycogenolysis is active
- Glycolysis is active
- Kreb's cycle and electron transport chain are both active, producing ample ATP
- **Fatty acid oxidation**
- ATP production is maximal
- Muscle fatty acid oxidation defect → Myopathic beta-oxidation defect → Inability to tolerate aerobic activity

Anaerobic/ strenuous/ ischemic exercise

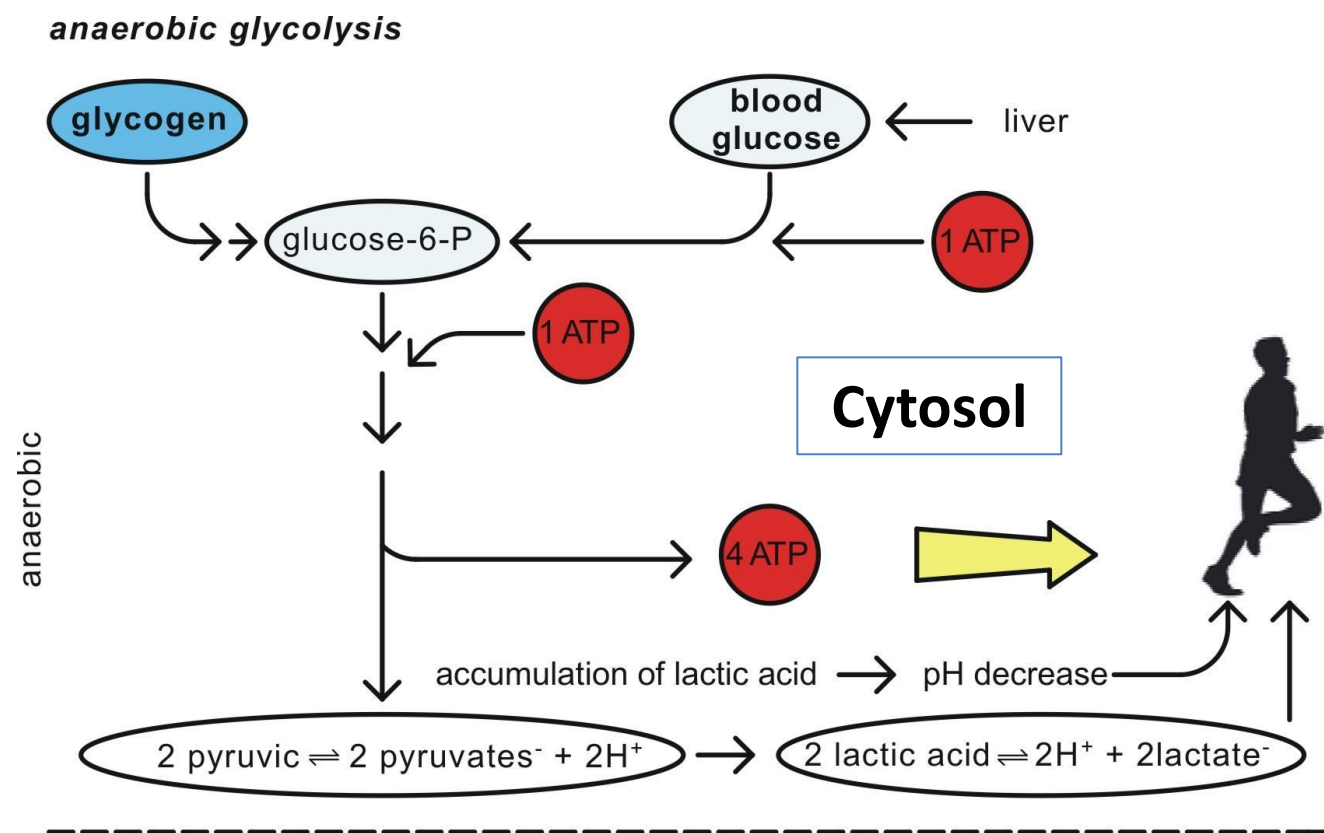


- Oxygen absent; Cytosol
- Glycogenolysis is active, but of short duration
- Glycolysis is active
- Pyruvate reduced to LACTATE
- ATP production is rapid but minimal
- Defect in Muscle Glycogenolysis/ glycolysis → Muscle glycogen storage disorder → Inability to tolerate anaerobic activity

From DM lecture 1

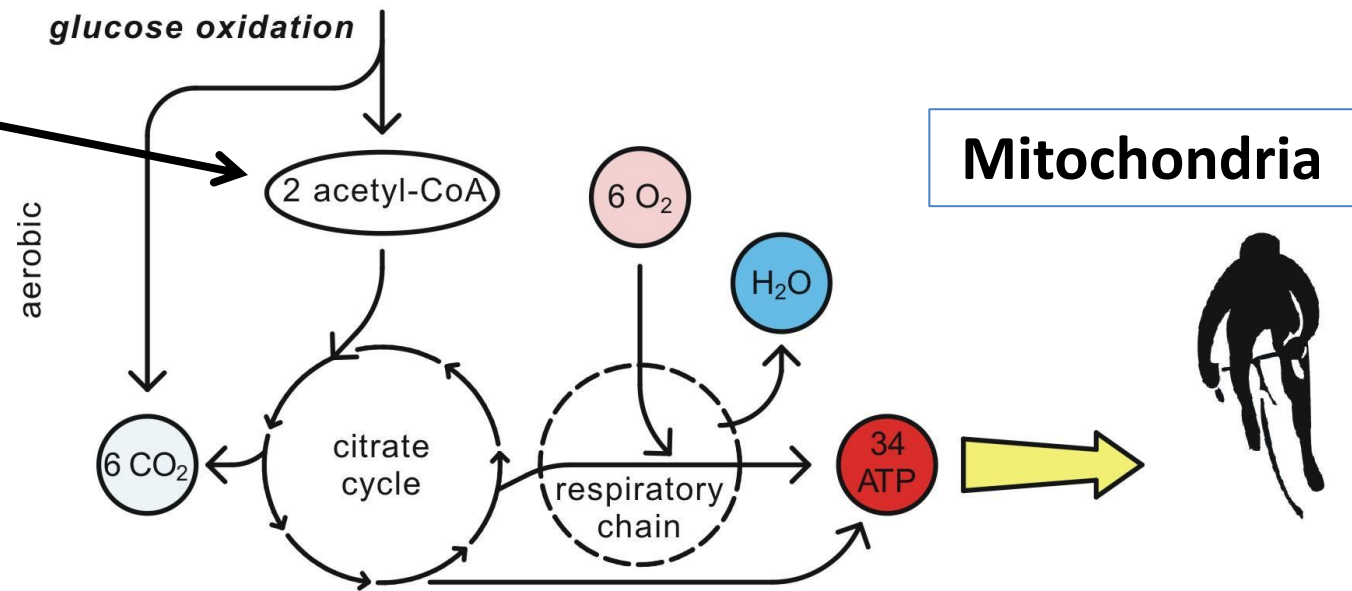
Metabolism in muscle: Aerobic and anaerobic

Glycogenolysis and Glycolysis (Anaerobic)



Fatty acid oxidation

Aerobic: Fatty acid oxidation and Krebs cycle



Fatty acid oxidation disorders

- **Systemic** fatty acid oxidation disorders
- **Age: 6 months-2 years**
- Organs affected: Liver and muscle
- Hypoglycemia and hypoketosis (metabolic crisis) following an illness
- Seen in: MCAD deficiency/ CPT-I deficiency/ primary carnitine deficiency
- Management: Prevent hypoglycemia; Carnitine supplements
- Differentiate from vonGierke disease

- **Myopathic** fatty acid oxidation disorders
- **Age: Adolescents/ Young adults**
- Organs affected: Muscle (Skeletal and cardiac)
- Muscle cramps during **aerobic exercise**
- Lab tests: Myoglobinuria and increased CK-MM levels
- Muscle biopsy: **Elevated muscle triacylglycerol (Lipid droplets in muscle)**
- Forearm exercise test: Lactate levels increase
- Management: Carnitine supplements
- Seen in: CPT-II deficiency
- Differentiate from McArdle disease



Jamaican vomiting sickness

- Eating unripe ackee fruit (Caribbean and Africa)
- Results in hypoglycemia and vomiting (2-6 hours)
- Contains '**hypoglycin A**' - Inhibits **MCAD (β -oxidation)**
- Inhibition of β -oxidation leads to profound **hypoglycemia**
 - Less acetyl CoA to activate gluconeogenesis (Absolute activator for Pyruvate carboxylase)
- Drowsiness (**Hypoglycemia**), coma and death
- Medium chain acylcarnitines found in urine

<https://www.facebook.com/StGeorgesU/videos/10153414290046461/>

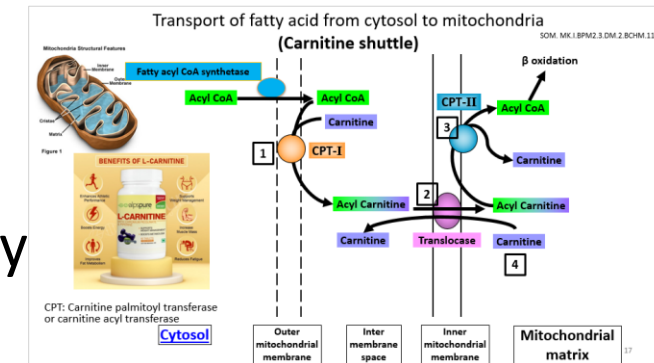
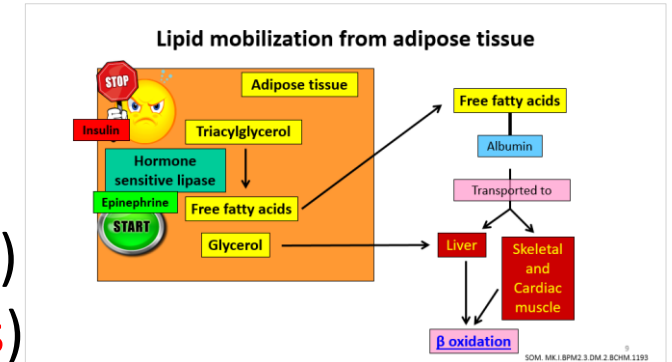


Unripe ackee fruit



Regulation of β -oxidation

- Serum free fatty acid levels regulate rate of fatty acid oxidation
 - Low insulin/ glucagon ratio (fasting; Type-1 diabetes mellitus) activates hormone sensitive lipase in adipose tissue (**lipolysis**)
 - High insulin/glucagon ratio (fed state) inhibits HS lipase
- Entry of fatty acids into mitochondria by CPT-I
 - Malonyl CoA (fatty acid synthesis) inhibits CPT-I
 - β -oxidation and fatty acid synthesis do not occur simultaneously (Reciprocal regulation)



Compare muscle glycogen storage disorder and myopathic fatty acid oxidation defect

Myopathic fatty acid oxidation defect

- Cramps during **aerobic exercise**
- Age: Adolescents/ Young adults
- Blood lactate levels increase following ischemic/ anaerobic exercise (normal test)
- **Muscle biopsy: Excessive lipid (TAG) droplets**
- Elevated serum CK-MM and myoglobinuria
- **CPT-II deficiency**

Muscle Glycogen storage disorder (McArdle disease)

- Cramps during **anaerobic exercise** (strenuous exercise)
- Age: Adolescents/ Young adults
- **NO increase in blood lactate following ischemic exercise**
- **Muscle biopsy: Excessive Glycogen accumulation**
- Elevated serum CK-MM and myoglobinuria
- McArdle: Muscle glycogen phosphorylase deficiency

Hormone sensitive lipase vs Lipoprotein lipase vs Pancreatic lipase

Hormone sensitive lipase

- Location: Within adipocytes
- Role of insulin: Insulin inhibits HS lipase
- Action: Breaks down storage TAG in adipose tissue to free fatty acids and glycerol
- Lab marker for increased HS lipase activity: Elevated serum free fatty acids
- Increased HS lipase activity seen in
 - Starvation
 - Insulin resistance (Metabolic syndrome)
 - Uncontrolled type 1 diabetes mellitus – HS lipase activity markedly increased

Lipoprotein lipase

- Location: Endothelium of blood vessels in adipose tissue and muscle
- Insulin induces lipoprotein lipase and required for optimum activity
- Action: breaks down TAG in lipoproteins (chylomicrons and VLDL) to free fatty acids that enter adipocytes and form TAG for **storage within adipocytes**
- Lab marker for low lipoprotein lipase activity: Increased TAG; and Increased VLDL
- Decreased activity of lipoprotein lipase is seen in
 - Insulin resistance (Metabolic syndrome)
 - type 2 diabetes mellitus – this results in increased serum TAG (VLDL) – which in turn increases the risk for cardiovascular disease (macrovascular complications)

Pancreatic lipase: Enzyme for digestion of dietary TAG

Comparison of synthesis and degradation of long chain even numbered fatty acids

	SYNTHESIS	DEGRADATION
Greatest flux through pathway	After carbohydrate-rich meal	In starvation
Hormonal state favoring pathway	High insulin/glucagon ratio	Low insulin/glucagon ratio
Major tissue site	Primarily liver	Muscle, liver
Subcellular location	Primarily cytosol	Primarily mitochondria
Carriers of acyl/acetyl groups between mitochondria and cytosol	Citrate (mitochondria to cytosol)	Carnitine (cytosol to mitochondria)
Phosphopantetheine-containing active carriers	Acyl carrier protein domain, coenzyme A	Coenzyme A
Oxidation/reduction coenzymers	NADPH (reduction)	NAD ⁺ , FAD (oxidation)
Two-carbon donor/product	Malonyl CoA: donor of one acetyl group	Acetyl CoA: product of β -oxidation
Activator	Citrate	
Inhibitor	Long-chain fatty acyl CoA (inhibits <i>acetyl CoA carboxylase</i>)	Malonyl CoA (inhibits <i>carnitine palmitoyltransferase-I</i>)
Product of pathway	Palmitate	Acetyl CoA
Repetitive four-step process	Condensation, reduction, dehydration, reduction	Dehydrogenation, hydration, dehydrogenation, thiolysis